Climate Change, Stock Productivity, and Fishing Effort in Chile's Multi-Species Small Pelagic Fishery

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Abstract

We study how climate change affects the long-run productivity of the three stocks in Chile's multi-species small pelagic fishery—anchoveta (*Engraulis ringens*), sardina común (*Strangomera bentincki*), and jack mackerel (*Trachurus murphyi*)—and the distributional implications for the artisanal and industrial fleets that target them. We identify a set of structural climate shifters ($\rho_i^{SST}, \rho_i^{CHL}$), interpretable as semi-elasticities of intrinsic stock productivity with respect to sea-surface temperature and log-chlorophyll-a anomalies, within a Bayesian state-space specification of stock dynamics calibrated on official IFOP and SPRFMO assessments for 2000–2024. We complement this structural model with a negative binomial model for annual fishing trips as a function of prices, vessel-level harvest allocations, weather, and regulatory closures, estimated separately for the artisanal and industrial fleets. Applying the identified shifters to CMIP6 projections from the IPSL-CM6A-LR model under SSP2-4.5 and SSP5-8.5 at mid-century (2041–2060) and end-of-century (2081–2100), we quantify the comparative-statics change in each stock's intrinsic productivity as a full posterior distribution. Anchoveta's median intrinsic productivity is projected to decline by 51% under SSP2-4.5 mid-century and by 89% under SSP5-8.5 end-of-century, with sardina común's posterior median declining by 90% and 100% respectively; in both cases the posterior probability of a negative response is effectively one. The jack mackerel shifters are not identified in the 2000–2024 sample, a finding consistent with the stock's transboundary SPRFMO management. Because the artisanal fleet's species portfolio is concentrated in the sardina–anchoveta pair, it bears a sharper long-run harvest-capacity decline than the industrial fleet, whose portfolio is more diversified. These results reverse the sign of the distributional asymmetry implied by reduced-form comparative statics and carry direct implications for the Chilean quota-allocation architecture under climate change.

1 Introduction

The distribution and abundance of marine resources are changing in response to environmental conditions such as global ocean warming (**?**). Climate change is projected to shift species distributions, reducing availability in some areas while increasing it in others, with direct consequences for fishing communities (**?**). Understanding how these changes will propagate to fishing activity is a prerequisite for designing effective adaptation policies. Adaptation responses by fishers are heterogeneous and depend on institutional, economic, and ecological context (**?**; **?**), which motivates empirical analyses at the local level.

The Chilean small pelagic fishery (SPF) provides a particularly informative setting. It is the largest fishery in the country, accounting for almost 94% of total Chilean landings in 2019 (**?**), and is dominated by three species—anchoveta (*Engraulis ringens*), sardine (*Strangomera bentincki*), and jack mackerel (*Trachurus murphyi*)—that share habitat and are coupled through both trophic and market linkages. Cheung et al. (**?**) project Chile among the top fishing nations in terms of declines in maximum catch potential by mid-century, with an estimated reduction of 6–13% under the SRES A1B scenario. Yet the Southeast Pacific remains one of the least studied regions regarding the economic impacts of climate change on fisheries (**?**), and, to our knowledge, no existing study for Chile combines multi-species stock dynamics with vessel-level effort responses under projected climate scenarios.[1]

A multi-species framework is essential in this setting because the three target species are ecologically and economically interdependent. Jack mackerel preys on anchoveta and sardine, so reductions in anchoveta or sardine availability may reflect not only direct environmental drivers but also changes in predation pressure (**?**; **?**). Together with a fleet structure in which the industrial and artisanal segments target different species, this generates cross-fleet spillovers through biology, quota allocation rules, and shared port infrastructure. Current management—which sets the TAC on a single-species basis—does not internalize these interdependencies, strengthening the case for a multi-species approach.

We contribute to the literature in three ways. First, we identify—rather than merely estimate—a set of structural climate shifters that modulate the intrinsic growth rate of each stock, within a Bayesian state-space specification of stock dynamics calibrated on official IFOP and SPRFMO assessments. The identified parameters are interpretable as semi-elasticities of stock productivity with respect to sea-surface temperature and log-chlorophyll anomalies, and are invariant to the joint historical distribution of biomass, harvest, and environmental conditions. Second, we couple this structural model with a negative binomial model for annual fishing trips estimated separately for the industrial and artisanal fleets, and we evaluate the identified shifters at CMIP6 projections (IPSL-CM6A-LR, SSP2-4.5 and SSP5-8.5) to obtain posterior distributions of the long-run comparative-

[1] For Chile, the only empirical study of fisher behavior using discrete choice modeling that we are aware of is @Pena-Torres2017-gn, which analyses how the El Niño–Southern Oscillation (ENSO) affects location choices in the jack mackerel fishery.

statics change in each stock's intrinsic productivity at mid-century (2041–2060) and end-of-century (2081–2100). Third, we show that the direction of the distributional asymmetry between the artisanal and industrial fleets is set by the differential exposure of each fleet's species portfolio to the identified shifters—not by reduced-form correlations between biomass and environmental conditions—which reverses the sign of the distributional asymmetry obtained from an unstructured comparative-statics reading of a reduced-form Seemingly Unrelated Regression benchmark. The analysis highlights heterogeneous vulnerability across fleet segments, with direct implications for how Chile's quota-allocation architecture should adapt.

2 The small pelagic fishery in Chile

The Chilean small pelagic fishery is structured into two latitudinal zones with distinct species composition and fleet activity. In the northern zone competition is largely between anchoveta and jack mackerel, while in the Central-South region (Regions V–X and XIV) all three species play a major role. The Central-South is therefore the most relevant setting for the study of species interactions and potential substitution within a multispecies management framework, and it is the focus of this paper.

The jack mackerel fishery was initially concentrated in northern Chile but shifted to Central-South Chile in the mid-1980s, where the main fishing grounds have traditionally operated within 50 nautical miles of the coast (**?**). Landings of the three species have historically been destined primarily for the reduction industry: between 1987 and 2004, approximately 85% of jack mackerel landings were processed into fishmeal and fish oil (**?**). Today, the main ports servicing the SPF include San Antonio, Tomé, Talcahuano, San Vicente, Coronel, Lota, and Corral. Sardine and anchoveta are harvested as a mixed fishery in the Central-South region: the two species share the same habitat and are caught with the same gear, making differentiation at the point of capture effectively impossible (**?**).

2.1 Stock status

Central-South anchoveta was classified as collapsed through 2018, shifted to overexploited status in 2019, and has been fished within maximum sustainable yield (MSY) limits since 2020. Sardine has generally remained within MSY limits, except in 2021 and 2023 when it was classified as overexploited. Jack mackerel was overexploited until 2018 and has since been harvested within MSY limits.

2.2 Management regime

The Chilean SPF is managed through an annual Total Allowable Catch (TAC, *Cuota Global*) divided between the industrial and artisanal sectors, with small shares reserved for research and

contingency. The TAC is subdivided by region and season, and unused quotas can be reassigned within the fishing year. Anchoveta and sardine are regulated as a mixed-species fishery—each species has its own quota, but substitution between them is permitted.

Since 2013, the industrial sector has operated under an individual transferable quota (ITQ) system known as Transferable Fishing Licenses (*Licencias Transables de Pesca*, LTP). Class A licenses were allocated on the basis of historical catches, while Class B licenses (up to 15% of the industrial fraction) are allocated through sealed-bid first-price auctions that began in 2015. Peña-Torres et al. (**?**) document that these auctions have exhibited low participation, limited ability to reflect economies of scale, and signs of coordinated bidding. The artisanal sector operates under a regulated freedom-to-fish regime, with regional TACs allocated through the *Régimen Artesanal de Extracción* (RAE) by area, vessel size, landing site, or fisher organization.

Biological closures (*vedas*) are a central regulatory instrument for sardine and anchoveta. In the Central-South region, reproductive closures run from July–October in regions IX–XIV and from August–October in regions V–VIII, while recruitment closures run from January to February across all regions. In total these yield 151 closed days per year in regions V–VIII and 182 closed days in regions IX–XIV. Jack mackerel has no seasonal biological closures.

3 Data and methodology

Our empirical approach has three stages. First, we identify a multi-species stock dynamics model using a Bayesian state-space specification on the latent biomass of each stock, with a Pella–Tomlinson surplus-production function whose intrinsic growth rate is modulated by an interpretable climate shifter. Second, we estimate a negative binomial model for annual fishing trips as a function of prices, harvest allocations, weather, and regulatory closures, separately for the artisanal and industrial fleets. Third, we evaluate the identified shifters under CMIP6 projections (IPSL-CM6A-LR) to assess how climate change will reshape the long-run productivity of each stock and, through the trip equation, the fleet-level allocation of fishing effort.

3.1 Historical data

We use data requested from the Chilean Fisheries Development Institute (Instituto de Fomento Pesquero, IFOP) covering the 2013–2024 period. The dataset includes trip-level microdata, which contain detailed records on vessel identifiers, departure and arrival times, vessel capacity, fleet and gear type, ports of departure and landing, fishery codes, haul timing and location, species composition, retained catch, and trip activity. In addition, we requested annual information on stock abundance and vessel landings by port, county, region, country, and species. Finally, we use ex-vessel prices, reported monthly or annually by port, county, region, country, and species. These

prices reflect those paid by processing plants to fishers at the point of first sale and are obtained through IFOP's landing surveys, which do not necessarily cover all market transactions.

For the environmental covariates, we use data from the E.U. Copernicus Marine Service Information, accessed through the Copernicus Marine Toolbox API. Salinity, sea surface temperature, and current speed and direction were obtained from the Global Ocean Physics Reanalysis (GLORYS12V1), which provides data at a 1/12° horizontal resolution with 50 vertical levels (**?**). Wind speed and direction at the surface were obtained from the Global Ocean Hourly Reprocessed Sea Surface Wind and Stress from Scatterometer and Model dataset, available at 0.125° horizontal spatial resolution and hourly frequency (**?**). Chlorophyll-a concentrations were obtained from the Global Ocean Colour dataset, which provides data at ~4 km horizontal resolution (**?**). All environmental data were retrieved daily (hourly in the case of winds) for the 2013–2024 period, covering the Chilean Exclusive Economic Zone (EEZ) between 32°S and 41°S (Figure 1)

Imputation of missing jack mackerel biomass To address the years in which no acoustic survey was conducted in south-central Chile—or in which insufficient information was available to produce a stock assessment—we imputed missing jack mackerel biomass values. Of the 25 years in the sample period (2000–2024), jack mackerel CS biomass is directly observed in 17 years; the remaining 8 years require imputation. The primary predictor is jack mackerel biomass from the Northern zone, estimated from independent acoustic surveys conducted since 2010 (correlation with observed CS biomass: $r = 0.82$, $N = 7$ overlapping years).

We estimated five alternative GLM specifications with a Gamma distribution and log link, ranging from a parsimonious model with Northern biomass as the sole predictor (2 parameters) to a saturated specification including Northern biomass, its square, the SPRFMO spawning biomass index, and their interaction (5 parameters). All models using Northern biomass are estimated with only 7 observations (years where both CS and Norte are simultaneously observed). The most complex specification exhibited severe overfitting (pseudo-$R^2 = 0.97$, but biologically implausible predictions requiring ad hoc censoring in two years). We therefore selected a parsimonious specification with Northern biomass and its square as predictors (3 parameters), which achieves an in-sample pseudo-$R^2 = 0.90$ and a leave-one-out cross-validation $R^2 = 0.78$, indicating acceptable out-of-sample predictive performance. For the single year where Northern biomass is also unavailable (2022), we use linear interpolation between adjacent observed values as a fallback.

The SPRFMO spawning biomass index, which covers the transzonal stock across the EEZ of Chile, Ecuador, and Peru, was also considered as a predictor. However, its correlation with observed CS biomass is low ($r = 0.11$, $N = 17$), reflecting the weak correspondence between regional stock dynamics and local abundance in south-central Chile. The interpolated series is used in subsequent estimation. To assess sensitivity to the imputation procedure, we report robustness checks using only directly observed jack mackerel biomass (Appendix **??**).

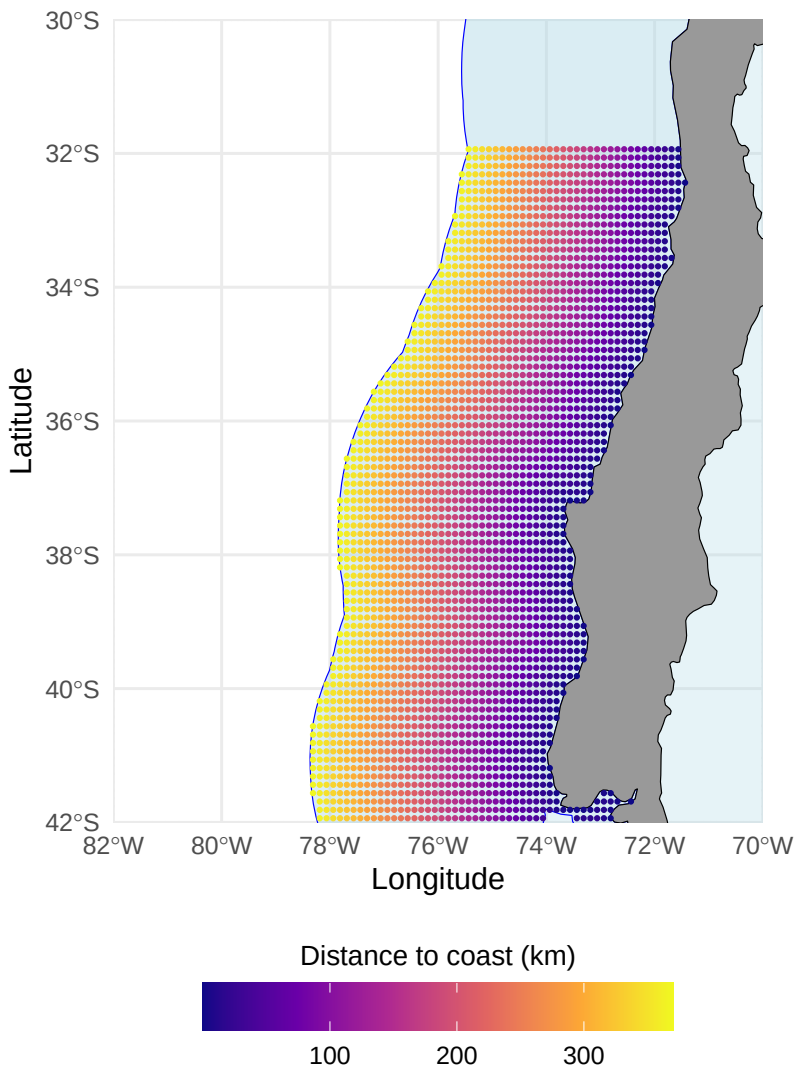

Figure 1: Geographical extent of the study area used for environmental covariates limited to the Chilean Exclusive Economic Zone

3.2 Future climate data

Climate projections for SST, chlorophyll-a, and near-surface wind speed are obtained from the CMIP6 archive, using the IPSL-CM6A-LR Earth system model. We use monthly outputs for two future scenarios—SSP2-4.5 (moderate) and SSP5-8.5 (high emissions). The projection horizons, baseline period, and delta-method procedure used to translate CMIP6 outputs into our empirical models are presented in Section 3.4.

3.3 Econometric models

3.3.1 Stock dynamics

We model the interannual dynamics of each stock $i \in \{\text{anchoveta}, \text{sardina común}, \text{jack mackerel}\}$ as a Bayesian state-space process on the latent biomass $B_{i,t}$, driven by a Pella–Tomlinson production function whose intrinsic growth rate is modulated by the physical climate. The law of motion is

$$B_{i,t+1} \;=\; B_{i,t} + r_{i,t}\, B_{i,t}\left[1 - \left(\frac{B_{i,t}}{K_i}\right)^{\theta_i}\right] \;-\; C_{i,t} \;+\; \varepsilon_{i,t}, \qquad \varepsilon_{i,t} \sim \mathcal{N}(0,\, \sigma^2_{\text{proc},i}), \tag{1}$$

where $C_{i,t}$ is the observed annual catch, K_i is the stock-specific carrying capacity, θ_i governs the asymmetry of the surplus-production curve (reducing to the Schaefer form at $\theta_i = 1$), and $\sigma_{\text{proc},i}$ is the standard deviation of the stochastic recruitment innovation. Climate enters the dynamics through a log-linear shifter on the intrinsic growth rate,

$$r_{i,t} \;=\; r_i^0 \,\exp\!\Big(\, \rho_i^{SST}\,(SST_{t-1} - \overline{SST}) \;+\; \rho_i^{CHL}\,(\log CHL_{t-1} - \overline{\log CHL})\,\Big), \tag{2}$$

so that ρ_i^{SST} and ρ_i^{CHL} are the semi-elasticities of stock productivity with respect to a one-degree Celsius SST anomaly and a one-log-point chlorophyll-a anomaly, respectively. The vector $(\rho_i^{SST}, \rho_i^{CHL})_{i=1}^3$ is the structural object on which Section 4.1 focuses. Because these parameters enter the law of motion as coefficients on exogenous climate forcings, they are invariant to the joint historical distribution of (SST, B, C) and therefore transportable to climate regimes outside the 2000–2024 estimation sample—a property that reduced-form correlations among growth increments and contemporaneous environmental conditions cannot satisfy by construction.

Latent biomass is linked to the official survey-based estimates of spawning stock biomass $B_{i,t}^{\text{obs}}$—published by IFOP for anchoveta and sardina común in Centro-Sur Chile and by SPRFMO for jack mackerel—through a log-normal measurement equation,

$$\log B_{i,t}^{\text{obs}} \;=\; \log B_{i,t} \;+\; u_{i,t}, \qquad u_{i,t} \sim \mathcal{N}(0,\, \sigma^2_{\text{obs},i}). \tag{3}$$

Environmental conditions are summarized annually by sea surface temperature (SST) and chlorophyll-a concentration (CHL), averaged over the Centro-Sur region within the Chilean EEZ and lagged one year to respect the pre-determined ordering of environmental forcing relative to year-t biomass. The choice of these two covariates follows a large biological literature that identifies thermal regime and primary productivity as the dominant local drivers of small-pelagic dynamics in the Humboldt Current System (**?**; **?**; **?**; see **?**; **?**).

The system is estimated by full-information Bayesian inference in Stan via Hamiltonian Monte Carlo. Priors on the biological parameters $(r_i^0, K_i, \theta_i, \sigma_{\text{proc},i}, \sigma_{\text{obs},i})$ are adopted from the single-

species official assessments of IFOP and SPRFMO; priors on the climate shifters $(\rho_i^{SST}, \rho_i^{CHL})$ are weakly informative normal densities centred at the point estimates of an independent stress-test regression of biomass growth increments on SST and log-CHL anomalies, with unit standard deviation (see Appendix A for full specification of the prior elicitation protocol). We estimate three nested specifications of increasing flexibility: (i) species-independent dynamics without climate shifters (`ind''), (ii) with a cross-stock residual covariance $\Omega$` (omega''), and (iii) the full model augmented with the climate shifter of Eq.~(2) (`full''). The posterior of the`full'' specification is our primary object of interpretation; the nested predictive comparison is relegated to Appendix B. All reported posteriors are based on four chains of 2{,}000 post-warmup iterations with standard convergence diagnostics ($\widehat{R} \leq 1.01$, bulk- and tail-ESS above 400 for all top-level parameters). The eight years in which the jack mackerel Centro-Sur survey was not conducted enter the likelihood as latent biomass states, so that the resulting posteriors of $\rho_{\text{jurel}}^{SST}, \rho_{\text{jurel}}^{CHL}$ propagate the associated uncertainty directly, superseding the Gamma-GLM imputation used by the reduced-form benchmark.

A Seemingly Unrelated Regression (SUR) benchmark on the same 2000–2024 sample, analogous in form to the specifications in Richter et al. (**?**) and Kasperski (**?**), is reported in Section~4.2 as a static correlational counterpart to Eq.~(1)–(2), with additional robustness panels in Appendix **??**. That benchmark is useful to anchor readers accustomed to reduced-form growth regressions, but the structural identification of Section 4.1 is the object on which the long-run comparative statics of Section 4.4 rely.

Figure 2 shows that anchoveta, sardina común, and jack mackerel exhibit distinct biomass trajectories over the 2000–2024 window, with no evidence of strong interannual co-movement across species once differences in scale are accounted for. Harvest pressure is visibly associated with these fluctuations; jack mackerel, in particular, experienced an abrupt decline during the late 2000s consistent with the combined effects of intense fishing and unfavorable environmental conditions.

The state-space formulation of Eq.~(1)–(3) accommodates both the measurement noise inherent in acoustic and egg-production surveys and the residual stochasticity of recruitment not captured by the climate shifter. Compared with a reduced-form SUR regression of growth increments on contemporaneous covariates, the state-space specification (i) separates process noise $\sigma_{\text{proc},i}$ from observation noise $\sigma_{\text{obs},i}$, (ii) respects the pre-determined ordering between year-$(t-1)$ environmental forcing and year-t biomass, and (iii) integrates out the missing jack mackerel observations within the likelihood rather than through a separate imputation step. The "omega'' specification preserves a free cross-stock residual covariance Ω as a robustness check against shared unobserved shocks; the comparison reported in Appendix~**??** shows that once climate shifters are introduced, the residual cross-stock covariance becomes statistically redundant (see Section 4.1).

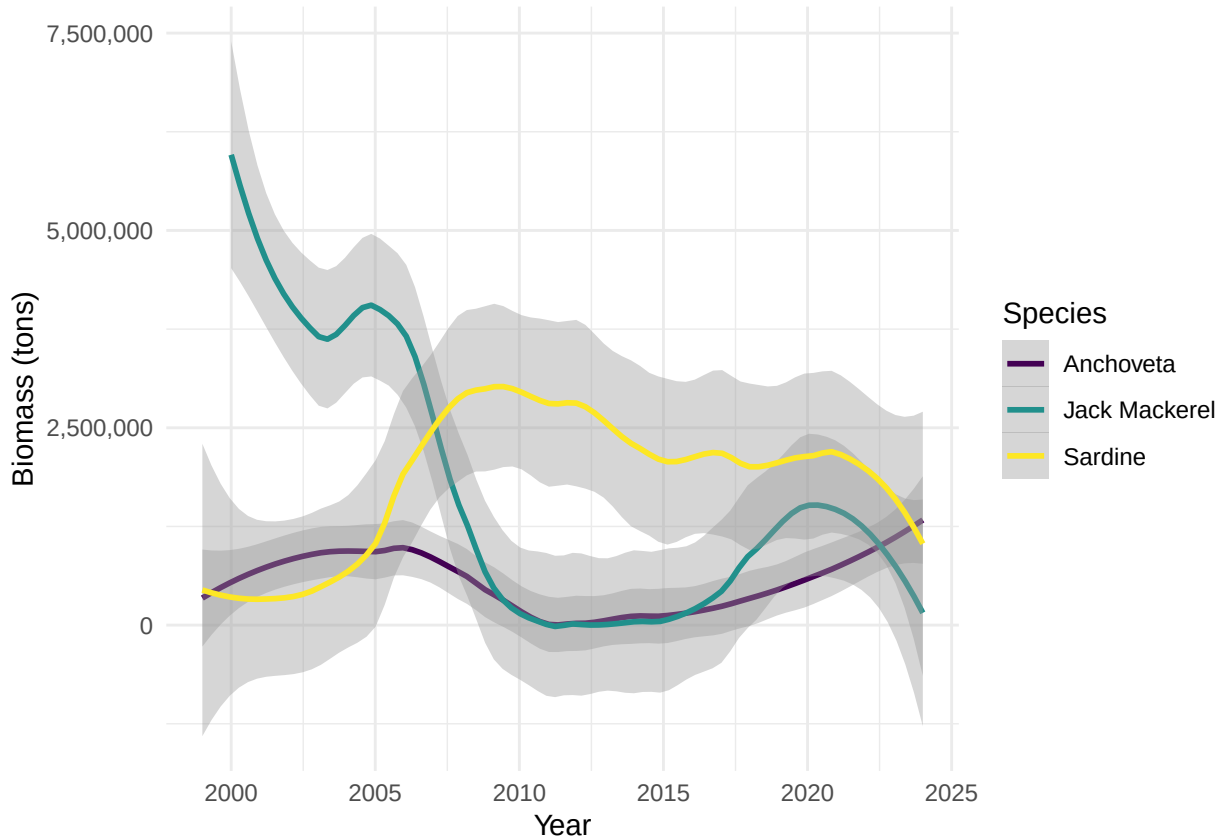

Figure 2: Estimated biomass of small pelagic species in Chile (2000–2024)

3.3.2 Total annual trips

We model the annual number of fishing trips taken by vessel v in year y as a count process following Kasperski (**?**). Since the available logbook data only record purse-seine operations, the unit of observation is a vessel–year. Separate effort models are estimated for the industrial and artisanal fleets, allowing fishing activity to respond differently to economic conditions and regulatory constraints across sectors. This specification explicitly accommodates technological heterogeneity across fleet segments, reflecting differences in vessel capacity, operating scale, and production technology.

The baseline specification is a Poisson model:

$$T_{vy} \sim \text{Poisson}(\lambda_{vy}), \qquad \lambda_{vy} = \exp\big(U'_{vy}\beta\big)\,, \tag{4}$$

where T_{vy} denotes the total number of purse-seine trips recorded for vessel v in year y. Given substantial overdispersion in the data—the variance-to-mean ratio of T_{vy} is 22.4 for the artisanal fleet and 4.9 for the industrial fleet—we estimate a negative binomial (NB) model, which nests the Poisson as a special case. A likelihood ratio test strongly rejects the Poisson restriction in both

fleets ($p < 0.001$).

The vector of explanatory variables U_{vy} includes output prices by species, vessel-level allocated harvest, fixed vessel characteristics, and operating conditions:

$$U_{vy} = [p_{sy},\, H_{vy}^{alloc},\, Z_v,\, O_{vy}]. \tag{5}$$

Output prices p_{sy} correspond to species-specific ex-vessel prices paid by processing plants to fishers, obtained from IFOP's manufacturing survey and deflated to constant 2018 pesos using the consumer price index from the Central Bank of Chile. Consistent with Dresdner et al. (**?**), prices are measured as annual averages over peak fishing months to reflect the economically relevant conditions faced by vessels when planning annual effort.

Because the objective of the model is to characterize vessels' effort responses to changes in prices, quotas, and environmental conditions within the small pelagic fishery, we restrict attention to SPF trips recorded in logbooks. These trips account for approximately 95% of observed fishing revenue and therefore capture the primary economic margin through which vessels adjust annual fishing effort.

Quota shares do not enter Eq.~(4) directly. Instead, shares translate annual TACs into vessel-level allocated harvests. Let ω_{vs}^r denote vessel v's historical share of landings of species s within its administrative region r (for artisanal vessels) or regulatory zone (for industrial vessels), computed over the full sample period. Given the effective TAC $\bar{Q}_{sy}^r$ for species s in year y in region r—obtained from SERNAPESCA quota monitoring records and reflecting all inter-fleet transfers and adjustments—vessel-level allocated harvest is:

$$H_{vy,s}^{alloc} = \omega_{vs}^r\, \bar{Q}_{sy}^r, \qquad H_{vy}^{alloc} = \sum_s H_{vy,s}^{alloc}. \tag{6}$$

Each vessel is assigned to its administrative region based on its modal departure port from the logbook records. Artisanal TACs are assigned at the regional level (regions V, VIII, IX, XIV, X), while industrial TACs follow the regulatory zone structure established by SUBPESCA (zones V–IX and XIV–X for jack mackerel; zone V–X for sardine and anchoveta). This regional construction introduces cross-sectional variation in H_{vy}^{alloc} that reflects the heterogeneous quota environments faced by vessels operating in different parts of the Centro-Sur fishery.

The vector Z_v contains time-invariant vessel characteristics, including hold capacity (log of cubic meters) and vessel type. Hold capacity determines the physical constraint on catch per trip and proxies for vessel scale.

The vector O_{vy} captures operating and regulatory constraints that vary across both vessels and years. Unlike the environmental variables in the stock dynamics model, which reflect biological productivity (SST, chlorophyll-a), the variables in the trip equation capture conditions that affect

the *feasibility* of fishing operations. Two vessel–year variables are constructed using each vessel's center of gravity (COG)—the catch-weighted centroid of its haul locations over the sample period. First, the number of days with adverse weather conditions is computed as the count of days per year in which wind speed exceeds 8~m/s at the environmental grid point nearest to the vessel's COG.[2] Second, the number of days closed due to biological closures (*vedas*) is assigned based on the regulatory zone corresponding to the vessel's COG latitude. In the Centro-Sur region, sardine and anchoveta face seasonal closures for reproduction (August–October in regions V–VIII; July–October in regions IX–XIV) and recruitment (January–February in all regions), yielding 151 and 182 closed days per year, respectively. Jack mackerel has no biological closures.

Quota prices are not included explicitly in the trip equation. Instead, the economic scarcity of quota is captured implicitly through aggregate TAC levels and vessel-level allocated harvest.

Trips, harvest, and prices may be jointly determined within the year, implying potential endogeneity in reduced-form trip regressions. However, the purpose of Eq.~(4) is not causal identification but rather to provide an empirically grounded behavioral relationship that maps changes in prices, quotas, and environmental conditions into expected fishing effort under projected climate scenarios.

3.4 Projection approach

We assess the impact of climate change on the Chilean small pelagic fishery through two channels: (i) a *direct weather channel*, in which projected changes in wind speed alter the number of bad-weather days that constrain fishing operations and thus enter the negative binomial trip equation; and (ii) an *indirect biomass channel*, in which projected changes in SST and CHL modify each stock's intrinsic productivity $r_{i,t}$ through the climate shifter $(\rho_i^{SST}, \rho_i^{CHL})$ of Eq.~(2) and, by that route, the law of motion in Eq.~(1).

Climate projections are derived from the IPSL-CM6A-LR Earth system model under two Shared Socioeconomic Pathways: SSP2-4.5 (moderate emissions) and SSP5-8.5 (high emissions). We apply the delta method (**?**; **?**), which preserves the observed interannual variability and fine-scale spatial structure from satellite records while imposing the climate change signal from CMIP6. For each variable, the delta is computed as the difference (additive for SST and wind speed) or ratio (multiplicative for chlorophyll-a) between the CMIP6 future climatology and the CMIP6 historical climatology (1995–2014 baseline), applied cyclically to each calendar month. We consider two future time windows: mid-century (2041–2060) and end-of-century (2081–2100).

For the direct weather channel, monthly wind speed deltas from CMIP6 are interpolated to the observed 0.125-degree wind grid and applied additively to daily wind speed observations. Bad-weather days (wind speed exceeding 8~m/s) are then recomputed under each scenario and entered

[2]The wind threshold of 8 m/s corresponds approximately to Beaufort scale 5, at which purse-seine operations become difficult for artisanal vessels. This threshold was selected based on model fit (AIC comparison across 8, 10, and 12 m/s thresholds); robustness checks are reported in Appendix **??**.

into the NB trip equation. For the indirect biomass channel, we evaluate the climate shifter of Eq.~(2) at the posterior draws of $(\rho_i^{SST}, \rho_i^{CHL})$ from the full state-space specification, taking ΔSST and $\Delta \log CHL$ as the scenario- and window-specific CMIP6 anomalies relative to the 2000–2024 estimation baseline. This yields a full posterior distribution over $r_{i,t}^{\star}/r_i^0$ under each climate regime—a long-run comparative-statics object, in the sense that it describes the steady-state productivity of each stock under a counterfactual climate, not a year-by-year forecast of biomass. Forward simulation of biomass and catch trajectories under endogenous harvest rules is carried out in a companion paper using the same posterior.

As noted by Kasperski (**?**), multi-species bioeconomic frameworks are well suited to long-horizon projections of fish populations rather than to intra-annual management responses. Because our empirical models are estimated on interannual variability rather than decadal trends, the projections reported here capture the mechanical effect of changed environmental conditions on stock productivity and trip counts while holding prices, quotas, and adaptive behaviour by fishers or managers fixed at historical levels (**?**). They provide a useful benchmark for the direction and magnitude of climate impacts, with the understanding that behavioural responses may attenuate or amplify these effects over time.

4 Results

4.1 Identification of climate shifters

The central econometric object of this paper is the vector of stock-specific *climate shifters* $\{\rho_i^{SST}, \rho_i^{CHL}\}_{i=1}^3$, which modulate the intrinsic growth rate of each stock as

$$r_{i,t} \;=\; r_i^0 \,\exp\!\big(\,\rho_i^{SST}\,(SST_{t-1} - \overline{SST}) \;+\; \rho_i^{CHL}\,(\log CHL_{t-1} - \overline{\log CHL})\,\big).$$

These shifters admit a clean economic reading: ρ_i^{SST} is the semi-elasticity of the stock's intrinsic productivity r_i with respect to a sea-surface temperature anomaly of one degree Celsius, and similarly ρ_i^{CHL} is the semi-elasticity with respect to a log-point of chlorophyll-a anomaly (a common proxy for primary productivity at the base of the food web). They are the structural parameters that couple the biological law of motion $B_{i,t+1} = B_{i,t} + g_i(B_{i,t}, X_t) - C_{i,t} + \varepsilon_{i,t}$ to the physical climate, and are therefore the quantities that must be identified for the long-run projections of Section 4.4 to have meaning as *comparative statics under climate change* rather than as out-of-sample forecasts.

We adopt weakly informative priors on $\rho_i^{SST}, \rho_i^{CHL}$ derived from independent reduced-form stress tests (Appendix A) and let the 2000–2024 biomass series update them. Table 1 reports the posterior means, 90% credible intervals, and the posterior-to-prior standard deviation ratio, which measures how much the data refine each parameter relative to the prior.

Table 1: Posterior identification of climate shifters.

Shifter	Stock	Prior $\mathcal{N}(\mu, \sigma)$	Posterior mean	Posterior 90% CI	$\sigma_{\text{post}}/\sigma_{\text{prior}}$	$\hat{R}$
ρ^{SST}	Anchoveta CS	$\mathcal{N}(-2.3, 1.0)$	-1.06	[-1.79, -0.38]	0.43	1.000
ρ^{SST}	Sardina comun CS	$\mathcal{N}(-2.0, 1.0)$	-2.75	[-3.46, -2.02]	0.44	1.000
ρ^{SST}	Jurel CS	$\mathcal{N}(0.0, 1.0)$	0.42	[-1.25, 2.09]	1.01	1.000
ρ^{CHL}	Anchoveta CS	$\mathcal{N}(-2.3, 1.0)$	-3.64	[-5.01, -2.25]	0.83	1.000
ρ^{CHL}	Sardina comun CS	$\mathcal{N}(2.1, 1.0)$	2.17	[0.96, 3.37]	0.74	1.000
ρ^{CHL}	Jurel CS	$\mathcal{N}(0.0, 1.0)$	-0.06	[-1.72, 1.60]	1.01	1.001

Note:
Shifters are the semi-elasticities of the intrinsic growth rate r_i with respect to SST anomalies (in °C) and log-CHL anomalies, respectively. Priors are adopted from the stock-specific stress-test results reported in Appendix A. The posterior-to-prior standard deviation ratio measures the amount of prior-to-posterior updating: values near one indicate the data carry little information about the parameter, whereas values well below one indicate identification from the likelihood. All R-hat 1.01.

The pattern of identification is itself informative and organises the economic interpretation of the paper. Three facts stand out.

First, for the two coastal upwelling stocks—anchoveta and sardina común—both climate shifters are identified with substantial precision. The posterior 90% credible intervals exclude zero for all four parameters, and the standard deviation ratios fall between 0.43 and 0.83, indicating that the 2000–2024 biomass series refines the priors materially. For anchoveta, a one-degree positive SST anomaly is associated with a semi-elasticity of $\rho^{SST}_{\text{anch}} = -1.06$ (CI $[-1.79, -0.38]$); for sardina común, the corresponding semi-elasticity is more than twice as large, $\rho^{SST}_{\text{sard}} = -2.75$ (CI $[-3.46, -2.02]$). Evaluated at one historical standard deviation of the SST anomaly series (estimation-sample $\hat{\sigma}_{SST} = 0.26\,^{\circ}\text{C}$ over 2000–2024), a thermal shock of that magnitude lowers the intrinsic growth rate of anchoveta by approximately 24% ($1 - e^{-1.06 \cdot 0.26}$) and that of sardina by approximately 51% ($1 - e^{-2.75 \cdot 0.26}$) relative to the climatological mean. Under a policy-relevant warming signal of $+1\,^{\circ}\text{C}$, comparable to mid-century CMIP6 SSP2-4.5 projections for the Chilean coastal zone, the implied declines are 65% for anchoveta and 94% for sardina, although $+1\,^{\circ}\text{C}$ lies approximately four historical standard deviations outside the estimation window and these projections therefore rely on the log-linear extrapolation embedded in the shifter function. Either way, the magnitudes are economically first-order for a fishery whose Centro-Sur TAC is in the low hundred-thousand-tonne range. The narrow historical SST variance ($\hat{\sigma}_{SST} = 0.26\,^{\circ}\text{C}$) also explains, mechanically, why the one-step-ahead forecasting test of Appendix B fails to discriminate between the climate-augmented and the climate-free specifications: within the estimation window the annual climate signal is small relative to the stochastic process innovation σ_{proc}, yet accumulates to economically consequential levels only over decadal horizons.

Second, the two coastal stocks respond to chlorophyll-a with **opposite signs**: $\rho^{CHL}_{\text{anch}} = -3.64$ (CI $[-5.01, -2.25]$) versus $\rho^{CHL}_{\text{sard}} = +2.17$ (CI $[+0.96, +3.37]$). At one historical standard deviation of $\log CHL$ ($\hat{\sigma}_{\log CHL} = 0.095$, a primary-productivity anomaly of roughly 10%), a positive CHL shock

lowers anchoveta growth by approximately 29% while raising sardina growth by approximately 23%. This is a reduced-form footprint of the well-documented trophic asymmetry between the two species: sardina filter-feeds on smaller phytoplankton and benefits from productivity pulses, while anchoveta preys on larger zooplankton whose populations are suppressed when the food web is dominated by small primary producers. The economic implication is first-order for this paper: the artisanal fleet, which lands both species, is not simply "climate-exposed"; it is exposed to **within-target heterogeneity** in the sign of climate effects. A warming-and-greening regime therefore does not translate into a uniform shock on the artisanal sector, but into a substitution across species that the current quota-allocation rules, indexed by species, cannot fully absorb.

Third, and in marked contrast, **neither SST nor CHL shifters are identified for jack mackerel**. The posterior standard deviations σ_{post} are indistinguishable from the prior scale ($\sigma_{\text{post}}/\sigma_{\text{prior}} \approx 1$), the posterior means lie near zero, and the credible intervals straddle zero in both directions. This is the expected result under the null hypothesis that jack mackerel productivity at the Centro-Sur scale is not locally coupled to the Chilean coastal upwelling regime—which is biologically sensible given its transboundary Southeast Pacific stock structure managed through SPRFMO. Econometrically, the model correctly reports non-identification rather than producing a spurious coefficient, which is a strength of the Bayesian state-space approach with structurally informed priors: the data can either move the prior or, as here, confirm that they are silent about it.

Figure 3 visualises the posterior–prior updating as a forest plot.

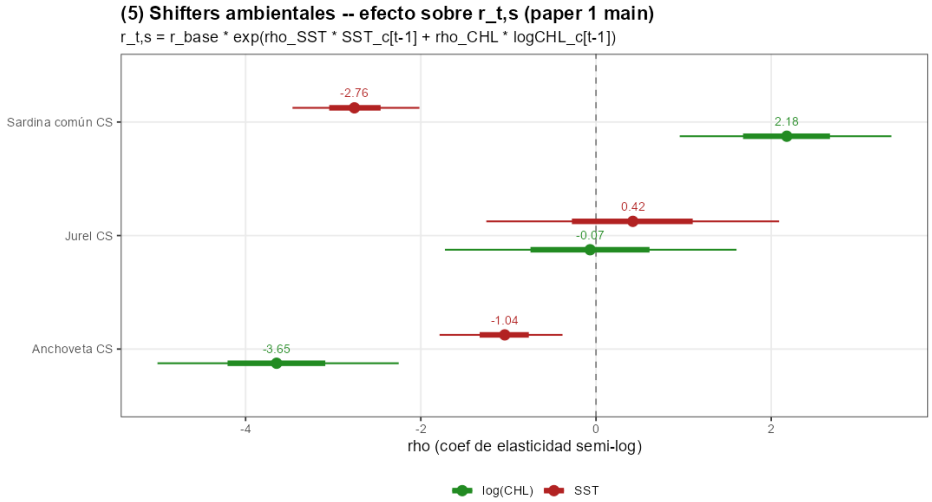

Figure 3: Posterior and prior distributions of the climate shifters $\rho_i^{SST}, \rho_i^{CHL}$ by stock. Anchoveta and sardina común show substantial posterior updating with narrow credible intervals excluding zero; jack mackerel posteriors coincide with the weakly informative prior centred at zero.

4.1.1 Identification versus short-run prediction

The identified shifters are structural parameters rather than reduced-form correlations: they measure an invariant biological mechanism linking ocean state to stock productivity, and they enter the law of motion as coefficients on exogenous climate inputs. This structural reading is what licenses their use as projection inputs under climate regimes outside the 2000–2024 estimation sample; reduced-form correlations, by contrast, would require the joint distribution of (SST, B, C) to remain stationary into the projection horizon, which is precisely the assumption the climate scenarios are designed to violate. Their economic value lies in **long-run comparative statics**: given a projected climate regime $X_t^\star$ for the period 2040–2100 drawn from the CMIP6 ensemble, the intrinsic productivity under that regime is $r_i^\star = r_i^0 \exp(\rho_i^{SST} \Delta SST_t + \rho_i^{CHL} \Delta \log CHL_t)$, and the expected steady-state biomass, harvest, and fleet-level effort follow mechanically. This is the channel through which the paper translates physical climate change into distributional outcomes in Section 4.4.

Crucially, identification of ρ does **not** imply superior short-run predictive power. A leave-future-out cross-validation at the one-year horizon (Appendix B) shows that the fully climate-augmented specification improves one-step-ahead log predictive density by only $\Delta\widehat{\text{ELPD}}_{\text{LFO}} \approx 0.4$ over a climate-free Schaefer baseline, a statistically unresolved difference. This is not a failure; it is precisely what should occur when the process-noise variance σ_{proc} is large relative to year-to-year climate innovations. Annual climate variability contributes little to *next-year nowcasts* but accumulates systematically over decadal horizons, and the shifters ρ_i are the parameters that quantify that accumulation. A robustness comparison using PSIS-LOO is reported in Appendix B (Table B.1) and confirms that the full specification is preferred as an in-sample description of the dynamics ($\Delta\widehat{\text{ELPD}}_{\text{LOO}} = 14.59$, SE $= 4.93$); the LOO and LFO results are complementary diagnostics of, respectively, identification-through-fit and forecasting-on-new-data, and the paper relies on the former for the economic interpretation.

4.2 Stock biomass model: SUR reduced-form benchmark

Table 2 summarizes the model selection exercise. Five candidate specifications were compared, ranging from a baseline with only own-biomass terms to a full model including quadratic chlorophyll-a and cross-species interaction terms. The preferred specification—which includes own biomass, its square, SST, SST^2, and chlorophyll-a—yields the lowest AIC and BIC. Adding quadratic CHL terms or cross-species interactions does not improve fit; in fact, both AIC and BIC increase when these terms are included. Wind variables were excluded in an earlier specification search, as a joint exclusion test failed to reject their omission. The preferred specification contains six parameters per equation, yielding approximately 17 residual degrees of freedom with $N = 23$ observations.

Table 3 reports the SUR estimates of annual biomass dynamics for sardine, anchoveta, and jack

mackerel. The system is estimated via robust maximum likelihood (MLR), yielding Huber–White standard errors and allowing unrestricted contemporaneous correlation across equations.

All three species exhibit strong interannual persistence and significant density dependence, consistent with discrete logistic growth. However, the species differ markedly in their sensitivity to environmental conditions.

Sardine biomass displays the strongest environmental response. The coefficient on chlorophyll-a is positive and highly significant ($\hat{\rho}_{CHL} = 80.99$, $p < 0.001$), indicating that years with higher primary productivity favor sardine growth. The quadratic SST term is also significant and positive ($\hat{\rho}_{SST^2} = 54.74$, $p = 0.001$), suggesting a convex response to temperature anomalies—sardine biomass increases more sharply under larger SST deviations from the mean. This pattern is consistent with evidence from the Humboldt Current System, where sardine tends to exhibit smoother population changes associated with warmer or transitional oceanographic regimes (**?**; **?**).

Anchoveta biomass is dominated by density dependence, with the strongest own-biomass coefficient ($\hat{r} = 1.11$, $p < 0.001$) and a highly significant quadratic term ($\hat{\eta} = -0.202$, $p < 0.001$). SST has a marginally significant negative effect ($p = 0.087$), but neither SST^2 nor CHL are significant. This asymmetric environmental response—where anchoveta reacts primarily to temperature rather than productivity—is consistent with the ecological literature. Anchoveta in the Humboldt Current is known to respond sharply to short-term cold, nutrient-rich conditions, dominating during cold phases of decadal-scale oceanographic variability (**?**; **?**; **?**).

Jack mackerel shows strong persistence ($\hat{r} = 1.06$, $p < 0.001$) but a distinct environmental signature. The quadratic SST term is large and highly significant ($\hat{\rho}_{SST^2} = -56.49$, $p = 0.001$), indicating a clear thermal optimum: biomass declines when temperatures deviate substantially from the long-run mean in either direction. Neither linear SST nor CHL are individually significant, suggesting that jack mackerel growth is less sensitive to annual fluctuations in upwelling productivity than sardine. This weaker coupling to coastal environmental variability is consistent with jack mackerel's broader spatial distribution, deeper foraging behavior, and transboundary stock structure within the Southeast Pacific (**?**).

These contrasting environmental responses have direct implications for climate projections. Under warming scenarios, sardine may benefit from increased thermal variability if accompanied by sustained productivity, whereas anchoveta could face reduced biomass if warming erodes the cold, nutrient-rich conditions it depends on. Jack mackerel, with its thermal optimum response, would be adversely affected by both warming and cooling extremes.

Cross-species interactions were tested but did not improve model fit (Table 2). The residual covariances across equations are also statistically insignificant, suggesting that shared unobserved shocks—beyond those captured by SST and CHL—do not generate strong contemporaneous comovement across species. This does not rule out ecological interactions (e.g., predator–prey dynamics between jack mackerel and small pelagics), but it indicates that such effects are not detectable

in the annual aggregate biomass data given the available sample size.

Appendix **??** presents three robustness checks. Table **??** restricts the jack mackerel series to observed (non-interpolated) years, reducing the sample to 12 observations and requiring a more parsimonious specification without SST^2. Table **??** estimates a two-species SUR for sardine and anchoveta only, treating jack mackerel biomass as an exogenous covariate. Table **??** augments the main specification with cross-species interaction terms, confirming that their coefficients are individually and jointly insignificant. The core findings—strong density dependence, positive CHL effects on sardine, and a thermal optimum for jack mackerel—are qualitatively robust across all specifications, lending confidence that the results are not driven by the interpolation procedure.

Table 2: Model selection for SUR biomass dynamics

Specification	K/eq	AIC	BIC	ΔAIC
No environment	3	490.7	507.7	13.1
Main (SST + SST2 + CHL)	6	477.6	504.9	0.0
Main + CHL2	7	482.9	513.6	5.3
Main + interactions	8	485.0	519.1	7.4
Full (interactions + CHL2)	10	490.5	528.0	12.9

Table 3: SUR estimates of biomass dynamics (main specification, N = 23)

	Dependent variable:		
	Sardine	Anchoveta	Jack mackerel
	(1)	(2)	(3)
Own biomass (x_{it})	0.438^{***} (0.113)	1.112^{***} (0.157)	1.057^{***} (0.137)
Own biomass^2 (x_{it}^2)	-0.021^{***} (0.005)	-0.202^{***} (0.047)	-0.017^{***} (0.005)
SST	-9.158^{*} (5.409)	-4.972^{*} (2.906)	5.923 (5.161)
SST^2	54.738^{***} (16.038)	−4.608 (7.203)	-56.485^{***} (17.413)
Chlorophyll-a	80.992^{***} (17.061)	−5.366 (11.155)	15.454 (24.835)
Constant	23.745^{***} (2.132)	12.096^{***} (1.639)	28.188^{***} (3.376)
Observations	23	23	23
R^2	0.707	0.608	0.790
Adjusted R^2	0.621	0.493	0.729
Residual Std. Error (df = 17)	7.308	3.917	9.589

Note: *p<0.1; **p<0.05; ***p<0.01.
Robust (Huber–White) SEs via MLR. Biomass in 10^5 tons.
Environmental variables centered at sample means.
Jack mackerel biomass interpolated via Gamma GLM on Northern biomass (see Section 3.1).

4.3 Total annual trips

Table 4 reports the negative binomial estimates separately for each fleet segment.

The coefficient on allocated harvest is positive and highly significant in both fleets, confirming that vessels with larger quota allocations undertake more trips per year. This variable provides the

Table 4: Negative binomial estimates of annual fishing trips (2013–2024, $N_{IND} = 319$, $N_{ART} = 4498$)

	Dependent variable:	
	Industrial	Artisanal
	(1)	(2)
Hold capacity (log m^3)	−0.037 (0.201)	0.193*** (0.056)
Allocated harvest (H_{vy}^{alloc}, tons)	0.00003*** (0.00001)	0.001*** (0.00003)
Price jack mackerel (1000s \$/ton)	0.005*** (0.002)	−0.002** (0.001)
Price sardine (1000s \$/ton)	−0.001 (0.002)	0.003*** (0.001)
Price anchoveta (1000s \$/ton)	−0.00001 (0.0003)	−0.0005*** (0.0002)
Bad weather days	−0.0001 (0.001)	−0.002*** (0.0004)
Veda days	−0.081*** (0.004)	0.009*** (0.002)
Vessel type (UNK)		−0.518*** (0.109)
Vessel type (BR)		−1.033*** (0.173)
Vessel type (BRV)		−2.456*** (0.189)
Vessel type (L)		−0.103 (0.150)
Vessel type (LM)	−0.982*** (0.077)	0.266* (0.143)
Constant	14.638*** (1.995)	1.230*** (0.446)
Observations	319	4,201

Note: *p<0.1; **p<0.05; ***p<0.01.
Vessel-clustered robust SEs. Prices in constant 2018 thousands of pesos/ton.
Allocated harvest = historical vessel share × lagged aggregate harvest.
LR test NB vs Poisson: Industrial $\chi^2 = 524.8$; Artisanal $\chi^2 = 49060.9$ (both $p < 0.001$).

key linkage between the trip equation and the simulation framework: changes in regional TACs translate into vessel-level effort adjustments through the quota share mechanism.

The two fleets differ markedly in their price responsiveness. For the industrial fleet, the jack mackerel price is positive and significant ($p < 0.01$), consistent with this fleet's primary orientation toward jack mackerel. Sardine and anchoveta prices have no detectable effect on industrial effort, reflecting the limited participation of industrial vessels in the sardine-anchoveta fishery. For the artisanal fleet, the sardine price is positive and significant ($p < 0.01$), consistent with sardine being the dominant target species for artisanal vessels in the Centro-Sur region. The coefficient on the anchoveta price is negative and significant. This counterintuitive sign likely reflects simultaneity: years of low anchoveta availability produce both higher prices (through the inverse demand channel) and fewer trips (through reduced catch opportunities), generating a negative reduced-form correlation. As noted above, the purpose of this equation is not causal identification but rather to provide an empirically grounded mapping for the simulation, where prices are determined endogenously by the inverse demand module.

Hold capacity is not significant for the industrial fleet, where vessels are relatively homogeneous in scale, but is positive and significant for the artisanal fleet. Among artisanal vessels, larger hold capacity is associated with more annual trips, consistent with larger vessels being more commercially active and better able to sustain operations across varying conditions.

The environmental and regulatory variables show distinct patterns across fleets. For the artisanal fleet, adverse weather days are strongly negative and significant ($p < 0.001$), indicating that years with more days of wind speeds above 8~m/s substantially reduce fishing effort for smaller vessels. For the industrial fleet, adverse weather is not significant, consistent with larger vessels being better equipped to operate under rough conditions. Biological closure days are strongly negative for the industrial fleet, representing the largest effect in the model, consistent with the industrial fleet's dependence on access during open seasons. For the artisanal fleet, the positive coefficient on closure days reflects the current limitation that this variable varies only across regulatory zones (151 days for regions V–VIII versus 182 days for regions IX–XIV) and therefore captures locational differences in fishing activity rather than the causal effect of closures.[3]

Vessel type dummies capture residual heterogeneity in fleet composition. Among artisanal vessels, larger vessel categories (L, BRV) are associated with substantially fewer trips relative to the baseline, reflecting differences in trip duration and operational patterns across vessel classes.

4.4 Climate change projections

The projection methodology is summarized in Section 3.4. Under each scenario and time window we translate CMIP6-derived environmental deltas into (i) changes in bad-weather days entering the trip equation (direct channel) and (ii) posterior draws of the stock-specific effective growth rate $r^{\star}_{i,t}$ obtained by evaluating the identified climate shifter of Eq.~(2) at the projected SST and log-CHL anomalies (indirect channel).

4.4.1 Projected environmental changes

Table 5 summarizes the projected environmental changes for the Centro-Sur study area. Sea surface temperature increases by 0.8–2.3°C depending on the scenario, with the strongest warming under SSP5-8.5 end-of-century. Chlorophyll-a declines moderately (2–7%), consistent with increased stratification reducing nutrient supply to the surface. Wind speed changes are negligible across all scenarios (< 0.1~m/s), reflecting the limited sensitivity of near-surface atmospheric circulation to greenhouse forcing in the IPSL model for this region.

Table 5: Projected environmental changes for Centro-Sur Chile (IPSL-CM6A-LR, delta method)

SSP	Window	SST (deg C)	CHL (ratio)	Wind (m/s)
SSP2-4.5	2081–2100	1.480	0.942	0.037
SSP2-4.5	2041–2060	0.811	0.971	0.060
SSP5-8.5	2081–2100	2.333	0.959	0.068
SSP5-8.5	2041–2060	0.970	0.988	-0.049

[3]Once year-specific closure dates from annual SUBPESCA resolutions are incorporated, providing within-zone temporal variation, this coefficient is expected to become negative.